

Project Proposal

Airline Reservation System

By

Muqaddas Raheem

Roll No: RBSIT-19-01

Session: 2019-2023

Supervisor: Mr. Asif Raza

**Department of Information Technology, Regional Institute
of Allied Health Sciences and Higher Education**



Table of content

Content	Page
Abstract.....	3
Background of the project.....	3
Problem of statement.....	3
Solution overview.....	4
Objective of the project.....	5
Scope of the Project.....	5
Methodology.....	5
Expected Outcomes.....	6
Conclusion.....	6

Abstract:

Airline reservation system is a web based application that provides the facility of online flight reservation system. The user can view flight timings, book and cancel ticket. System also allows booking till the last minute as emergency services. To develop this project I have selected the “Waterfall Methodology” because this methodology fulfills all the requirements to create an acceptable application. Our system is one of the best opportunities for those who cannot afford enough time to get their tickets reserved standing in long queues. The system validates user information, processes payments securely, and updates the flight and user databases. Customers can also cancel reservations. This system minimizes manual errors in flight reservations, streamlining the booking, modification, and cancellation processes for customers.

Background of the Project

Airline reservation systems are like super-smart computers for booking flights. In the past, people had to visit ticket counters to book a seat on a plane, which was time-consuming. These systems used to rely on paperwork, making errors common. Thanks to technology, we now have automated systems. You can book flights online or through mobile apps, which is quick and easy. These systems also help airlines manage their flights efficiently and keep track of passengers. Imagine if you could book a flight as easily as ordering something online, choose your seat easily, and get all your travel information in one place. That's what we want to make happen. So, I am working on creating a new, modern system that will make booking flights a breeze. It will be simple to use, work on your computer and phone, and even help airlines run more smoothly. Overall, airline reservation systems have made travel more convenient, reducing the hassle of booking and improving the entire flying experience for everyone.

Problem Statement:

Every Airline company is charged with the responsibility of ensuring to give safe and comfortable service for its customers. To ensure this quality of service, the companies which work on this business should reach to their customers and give a quality service to compete in the market. And this includes easy ticket reservation and cancellation system. In the manual system, the daily reservation list for all flights is generated and sent. Very often, they cannot put up the reservation list on time. Owing to the inefficiencies in the manual system, when passengers cancel tickets, the reservation list is not updated in time.

They maintain a waiting list, which is used to update the reservation list when passengers cancel tickets. Currently, the manual system handles all requests for changes in reservation. It treats a change of this kind as a cancellation, and reissues tickets. Reservation opens few days before the scheduled departure date. Based on the availability of seats the tickets are issued. Each ticket, whether confirmed or wait-listed, has a unique identification number.

This number is generated in a serial order. The reservation clerk records the amount of fare paid for the ticket in Cash Collection Register. A passenger can cancel tickets by submitting a cancellation

form with the ticket. Depending on the difference in hours between the departure and cancellation the passenger loses certain percentage of the fare.

Limitation of the manual system

The limitations of the manual system are

1. Only few bookings could be made due to manual operation.
2. It takes enormous amount of time for recording transactions.
3. Requires a large amount of manual laborers.
4. Customers should go to ticket office store serve a ticket and cancel it.

Proposed system

In order to overcome the draw backs in the manual system, it was decided to create a system, which is

- Integrated
- Reliable
- Flexible
- Secure
- Consistent

This system helps in speedy information retrieval. Responds to various queries made at the different levels of management. The system is Menu driven and user friendly with attractive html interface. It assists the user with respective messages to overcome the errors at runtime. The system is designed in such a way that the reservation list is available on time before the scheduled departure time. The reservation list reflects the latest status with respect to the cancellation of tickets that helps the wait-listed.

Solution Overview:

Automating Booking: We plan to move from a manual process of booking flights to an automated system.

Online Access: Passengers will have the convenience of booking flights online from their computers or smartphones.

Efficient Management: The new system will help airlines efficiently manage flight reservations, reducing paperwork.

Secure Data: Passenger information will be stored securely in the digital system.

Real-Time Updates: Passengers can receive real-time updates on their flights, making travel planning smoother.

Streamlined Check-In: Passengers will have the option for online check-in, saving time at the airport.

All-in-One Place: All the important travel info, like flight details and boarding passes, will be in one convenient place passenger in knowing their present status. In the new system customers could reserve and cancel tickets from anywhere which have an internet access.

Objective of the project:

General Objective:

The main goal of the proposed system is to make online the processes of the air ticket reservation and cancelation system.

Specific Objectives:

Some of the Specific Objectives of the system are listed below:

- To create a distributed system that will be used by customers.
- To satisfy customers need by reaching them at their place.
- Provide large number of ticket reservation and cancel a service in a few time.
- To create a system that makes booking and cancelling easy and user friendly.
- Make the entire process of itinerary decisions On-The-Go.
- To provide the customer more options like last minute emergency booking, low-cost chartered flights etc.
- Reduce the work load of ticket officers and counters.

Scope of the project

This project designs and implements air ticket reservation system. Supported by a well-designed database, all available air flight information is integrated together and can be accessed easily through a single point. A friendly user interface is provided so that various combinations of search criteria can be fetched from user and generates corresponding database search statements. The air ticket reservation system provided both customer and administration interfaces with the latter used for administration purposes. If time permits, the reservation system will support frequent user registration and personal information management. It increases security, speed, storing and accuracy. It provides convenience to travelers. It decreases manpower and high cost.

Methodology:

A well-structured and maintained methodology can guide the developer to its best destination of application development. Adopting of the right and method for software development is very important. All the work plan and development process depends upon the method which the developer adopted. So, suitable and relevant method for the requirements, specification, constraints and many other aspects of software development is necessary. My adopted methodology is the “**Waterfall Methodology**”, the first and basic method of creating software.

Reason for Chosen Methodology

5 to 8 stages of software development are commonly identified by the people. Every model for software development should fulfill these stages requirements to create an acceptable application. If, these stages are not accomplished properly than there are much possibilities for failure of application. These stages are as follows.

- **Requirement Analysis and Definition**
- **System Software Design**
- **Implementation and Unit Testing**
- **Integration and System Testing**
- **Operation and Maintenance**

“Waterfall methodology” is phase developing methodology. Each phase is documented briefly and sends to the user for approval, if it is approved then the next phase is started. If the first phase is not approved by client or it does not according to the requirement of the client it is re-documented. Because my project requires the same situation so “Waterfall Methodology” is best to use.

Fact-finding Technique

- Questioner
- Interview
- Literature review

Development Tools

Activities	Tools/Programs
Client side coding	HTML
Client side scripting	JavaScript
Platform	Windows
Database server	MySQL
Web server	Xampp server
Server-side scripting	php
Browsers	Google chrome
Documentation	MS Word, Adobe reader

Table2

Expected Outcomes:

- Booking flights will become much easier and faster.
- Passengers will be able to book their flights online using their computers or smartphones.
- Airlines will have more efficient ways to manage flight reservations.
- Passenger information will be stored securely in a digital system.
- Passengers will receive real-time updates about their flights, making travel planning smoother.
- Online check-in options will save time at the airport.

This project aims to replace the old manual system with a new automated one, making the booking process quicker and more convenient for passengers while improving efficiency and security for airlines.

Conclusion:

"So, to sum it up, this project is all about making air travel easier for everyone. We want to move away from the old, time-consuming manual system and bring in a new, automated way of booking flights. When we do this, passengers will find it much simpler to book their flights online, airlines can manage things more efficiently, and overall, travel will be smoother and more convenient. It's a positive change that will benefit everyone involved - passengers and airlines alike. We're excited to make this happen and look forward to the positive impact it will have on air travel.

